

Assignment 1: Part C Big Data Analysis

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Student Name : Araju Nepal

Student Id: a1910508

Problem description data pre-processing

The input dataset consists of both quantitative and categorical variables related to females aged 15 - 45.

The key features in the input dataset are:

Quantitative variables	Age (yrs), Height (cm), Weight (Kg), Cycle length (days)
Categorical variables	Blood Group, BMI, Regular Cycle, Weight gain, Hair loss/baldness, Excessive body/facial hair, Skin darkening, Pimples on face/jawline, Regular fast food intake, Regular physical activity

The output data is PCOS, a categorical variable indicating if the female has Polycystic Ovary Syndrome (PCOS) or not.

Exploratory data analysis reveals that age, weight, height, and cycle length show diverse distributions, with a skew towards younger ages. It indicates PCOS associations with Blood Groups O+ and B+ and obesity. PCOS prevalence in Blood Groups O+ and B+, and strong associations with obesity, irregular cycles, weight gain, hair loss, excessive body/facial hair, skin darkening, and high fast food intake were confirmed from chi-square and point-biserial correlations.

The datasets were initially processed by standardizing column names, renaming for consistency, and merging into a unified dataset. Irrelevant columns were dropped, and variable names were standardized. Missing values for blood group, weight, height, and BMI were addressed: blood group was imputed based on overall distribution, BMI was computed from weight and height, and missing weight and height were imputed using mean values by BMI categories to maintain their distribution. Duplicate rows were removed, and outliers were identified and removed using boxplot visualization. No scaling was necessary as variables were comparable. This preprocessing ensured data uniformity and completeness for analyzing PCOS susceptibility in females aged 15-45.

Model selection

- Logistic Regression: This supervised ML algorithm predicts PCOS probability using a logistic function, based on various features. This model was chosen because it is a simple and interpretable model that quantifies relationships between predictors and outcomes, and is suited for analyzing how the features combined or independently influence PCOS susceptibility (Kanade 2022).
- Random forests: This model combines predictions from multiple decision trees to boost accuracy and generalize effectively. This model was chosen as it handles complex relationships among age, BMI, lifestyle, and blood type, while also preventing overfitting, dealing well with noisy data, and ranking feature importance to better predict nuanced patterns in diverse datasets (Wood 2019).

Model refinement:

The logistic regression model was refined using Exhaustive Feature Selection (EFS), which evaluates various feature combinations to identify the optimal set for improved accuracy (www.stratascratch.com n.d.). This involved evaluating every possible subset of features, to identify the subset with highest accuracy. Then, both initial and refined models were assessed through 5-fold cross-validation on the training set, measuring accuracy, precision, recall, and F1 score, to ensure a comprehensive evaluation.

The random forest classifier was initialized with default parameters and trained after encoding categorical variables. Hyperparameter tuning was performed through GridSearchCV (Koehrsen 2018). This involved an exhaustive search over parameters such as the number of estimators, maximum features, maximum depth, minimum samples to split, and minimum samples per leaf, all assessed via 5-fold cross-validation. The best model was selected based on these optimized parameters, and its feature importances were analyzed, then was evaluated on a test set.

These refinements ensure a robust evaluation, avoiding overfitting and unfair feature exclusion.

Performance measure

Performance measures such as accuracy, precision, recall, F1 score, and Area Under the ROC Curve (AUC) offer a comprehensive evaluation (Blog 2023). These metrics were selected for the holistic assessment.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	AUC
Logistic Regression	85.05	80.45	65.56	72.24	0.8856
Random Forest	94.84	96.27	85.93	90.80	0.9907

Result interpretation

The performance metrics of both models indicate that random forest outperforms logistic regression in several key areas. While logistic regression showed respectable results with cross-validation, random forest model achieved a higher overall metrics. Additionally, the random forest model's AUC surpasses the logistic regression model's AUC, indicating better overall classification performance. These metrics highlight the random forest model's superior capability in accurately predicting the likelihood of PCOS.

Feature importance analysis further supports the selection of random forest model. While both models identified significant predictors such as age, excessive body/facial hair, weight gain, and skin darkening, the random forest model was able to better capture the nuances of these predictors due to its ability to handle complex, non-linear relationships. Blood group features were found to have relatively low importance in both models, with B+ and O+ showing higher prevalence rates but not significantly influencing the outcome.

Therefore, the random forest model proves to be more effective in accurately predicting PCOS in females aged 15-45 with different blood groups. It stands out as the superior model due to its enhanced performance and robust handling of various influential factors.

References:

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